

National Land Acquisition & Management System

Smart India Hackathon (SIH) 2026 Technical Report

MEMBER PROFILE: Member 3 (Shivam)

Technical Role: Workflow State Machine, Validation & ML Specialist

Primary Focus: FSM Workflow State Transitions logic enforcement (Proposal -> GIS Verification -> Sec 11 -> Award)

1. Functional Domain & Work Done

As part of the NLAMS development team, Member 3 (Shivam) has taken ownership of the functional module targeting Workflow State Machine, Validation & ML Specialist. The following components have been fully implemented and integrated into the live workspace:

- **Real Machine Learning Pipeline:** Created train_model.py to fit Scikit-Learn Decision Trees on SQLite dataset inputs, outputting delay risks and completion forecasts.
- **Model Weights Serialization:** Dumped trained estimators to land_delay_model.joblib for REST API query loading.
- **Predict API Endpoint:** Added GET /projects/{id}/predict router in FastAPI, fetching real ML delays predictions.

2. Overall Website Layout & Features

The NLAMS portal operates as a single source of truth under a modernised, high-contrast, official government light theme (saffron, navy, and emerald accents). Key features include:

- **Tricolor Home & Directives:** An entry portal showcasing the LARR Act 2013 roadmap and operational pillars.
- **Bilingual Language Switcher:** A global toggle translating all KPI labels, timelines, Gazette notifications, and Treasury advice receipts dynamically into Hindi and English.
- **Executive Dashboard & GIS:** Renders land parcels as Leaflet map overlay polygons, color-coded by acquisition stage, alongside Recharts predictive dashboards.
- **Automated LARR Workflows:** A pipeline allowing Central, State, and District roles to sign off on stages, instantly generating printable official Gazette templates.
- **Web3 Trust & Compensation Portal:** Simulates smart contract escrow releases with Metamask wallet connectivity, a digital deed SHA-256 validation scanner, and a scrolling block ledger.
- **Mobile-Responsive Field Survey Node:** Allows field surveyors to capture coordinate coordinates using GPS triangulation and report soil classification, structures, and site images.
- **SMS Alert Simulator:** Fires dynamic, floating mobile network text notification alerts during proposal submissions, boundary checks, and DBT escrow releases.

3. Sequential Learning Pathway & Project Roadmap

To ensure a smooth transition from prototype to production for the SIH finals, the following learning benchmarks are mapped for this role:

- **Focus Area:** FSM Workflow State Transitions logic enforcement (Proposal -> GIS Verification -> Sec 11 -> Award -> Handover)
- **Focus Area:** Cross-field backend business logic validations
- **Focus Area:** Pandas data preprocessing, cleaning, and features encoding
- **Focus Area:** Scikit-Learn Classifier & Regressor training for project latency forecasting

4. Judge Defense Recommendations

During the pitch and technical Q&A, the judges will test the integrity of each module. Use the following defense arguments to demonstrate technical authority:

- **Defense Tip:** Explain that the Random Forest/Decision Tree model was trained using the 2,501 SQLite records on features (State, Area, Budget) and optimizes Recall to ensure bottlenecks are flagged early.